Hydrogen Sticking Dynamics on Graphene Surface using Hybrid Trajectories

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Abstract: The H sticking on graphene is to be investigated using Hybrid TRAJectory (HyTRAJ), trajectory-based quantum method. A single potential with well-defined barrier, physisorption and chemisorption well is to be obtained and the study of phonon vibrational modes using PWSCF method in Quantum Espresso. In this method, we have included the multi-phonon exchange, and we are in process of doing the dynamics.

Introduction

H Sticking Dynamics: Understanding the sticking of H on graphene is important for low energy interstellar medium where a physiosorbed H would be chemisorbed through quantum tunnelling.

- **Quantum Methods** (with T)
  - Morisset *et al.* [1] (DFT+CCWP method)
  - Karlicky *et al.* [2] (Chemisorption study using system bath model)
  - Lepetit *et al.* [3,4] (Physisorption study with several methods)
- **Mixed Quantum Classical** (with T)
  - Morisset *et al.* [5] (Classical Phonons + H motion 1D WP)

Challenges/Limitations
- Very Long
- One Phonon approximation
- Number of phonons at asymmetric points are limited
- Not very efficient to study molecules

- **MCTDH**: Bonfanti *et al.* [6,7,8] (T=0K)

Objective: To study the sticking of H atom on graphenic surfaces using a new dynamics method developed in the ANR HyTRAJ (a hybrid quantum trajectories method) including phonon in dynamics.

Sticking at collision energy < 0.2eV with surface temperature between 10K to 100K. Experiment by Zecho *et al.* [9] on graphite and Haberer *et al.* [10] on Graphene observe this at 150K-300K.

Theory:

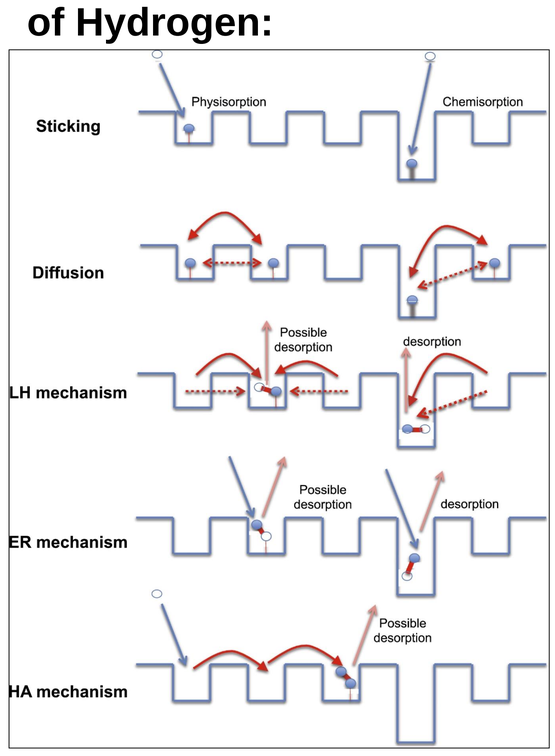
$$\hat{H} = \hat{T} + V(z,Q) \qquad \Psi(z,Q) = \sum_n c_n \phi_0(z)\phi_n(Q_n)$$

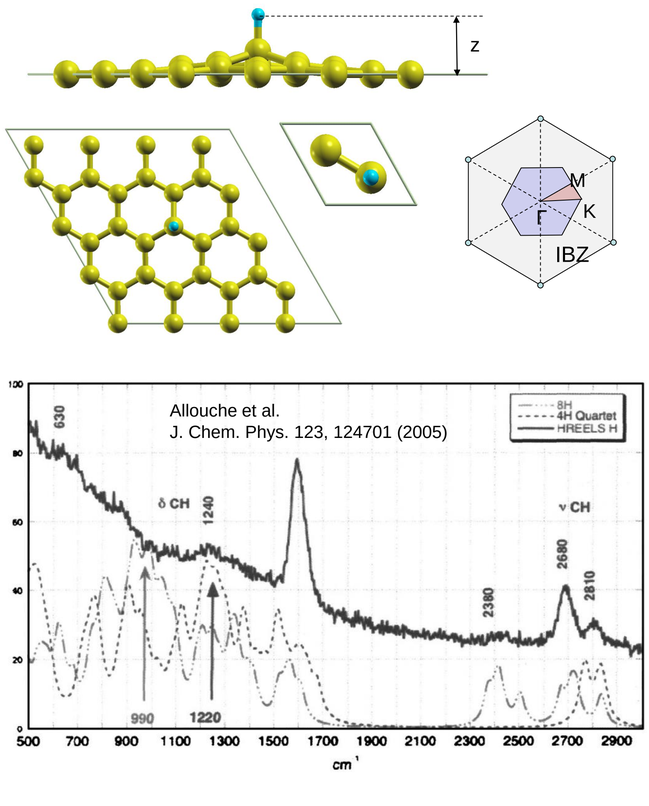
$$V(z,Q) = V_0(z) + \frac{1}{2}\sum_{ij} h_{ij}(z)(Q^i - Q_0^i(z))$$

- TDSE for dynamics using wave packet propagation

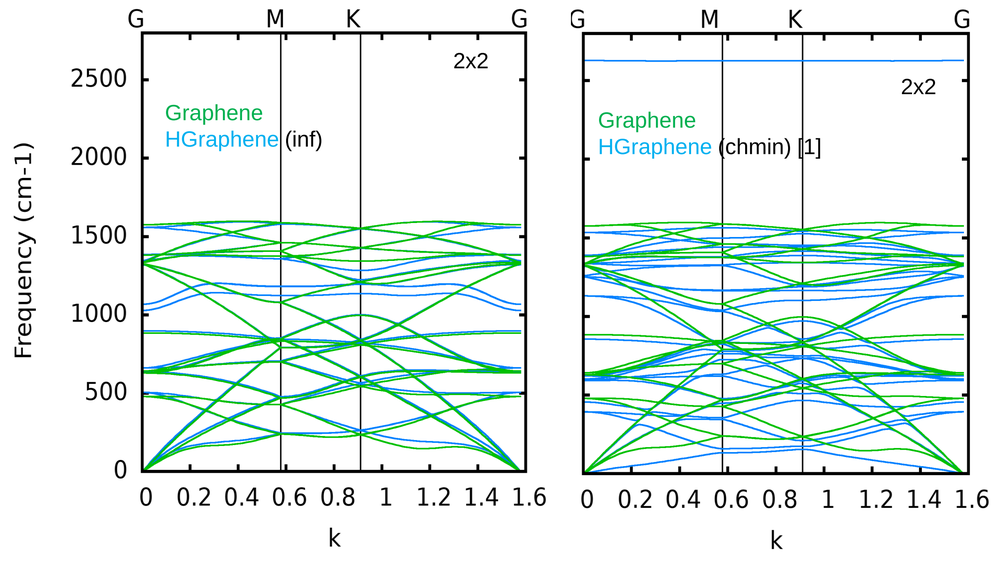
Essential Ingredients: Single PEC without sharp crossing, Phonon frequencies at Γ point and Phonon dispersion

Sticking and diffusion of Hydrogen:



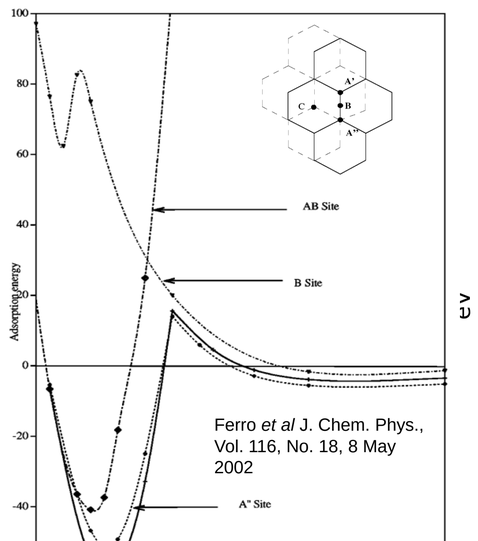


Dispersion plots :

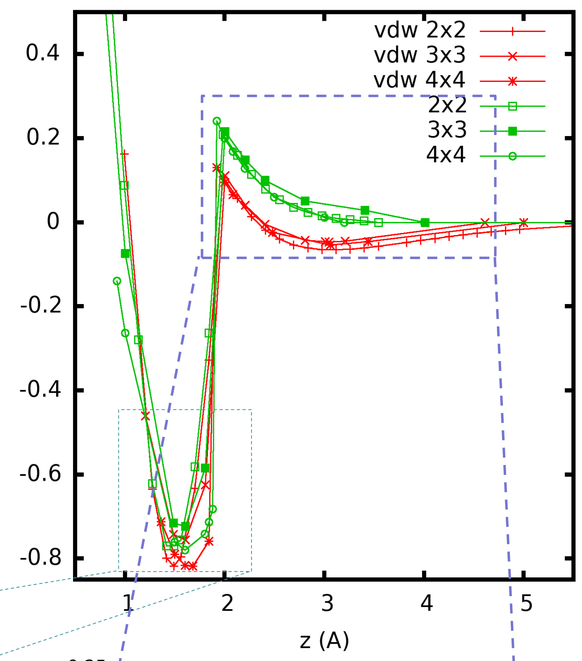


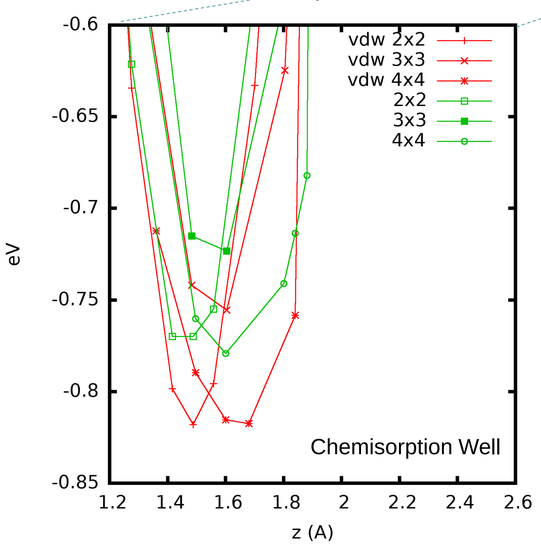
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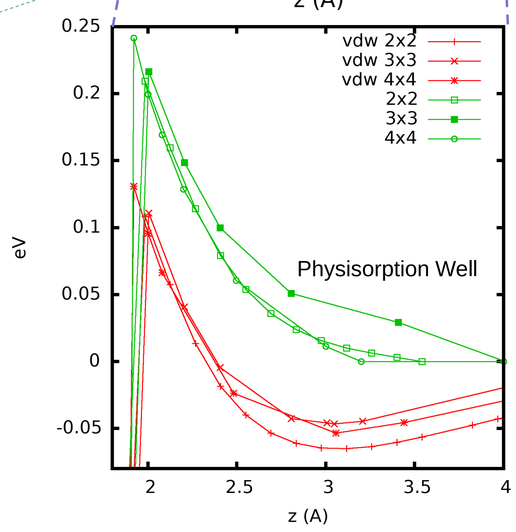
DFT Potential Energy Curve



Ferro *et al* J. Chem. Phys., Vol. 116, No. 18, 8 May 2002



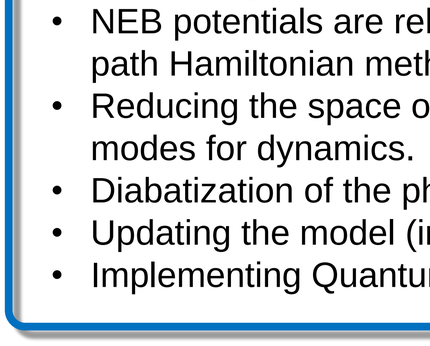


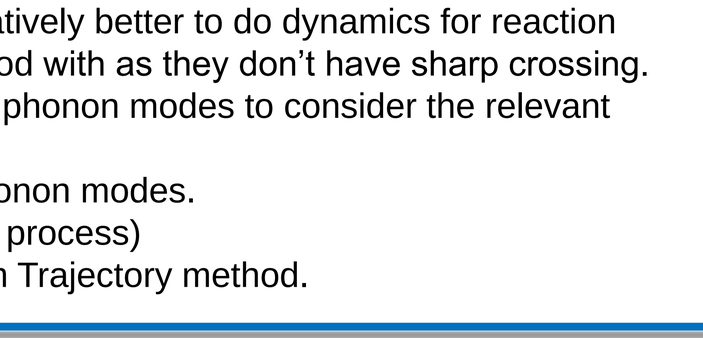


Ivanovskaya *et al.* / Rougeau *et al.* / Wang *et al.*

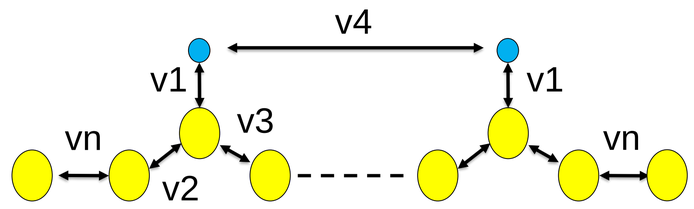
NEB calculation

Potential plots: 2x2





Vibrational Modes



Conclusion and Further Work

- Phonon calculation using Quantum Espresso are very time consuming.
- NEB potentials are relatively better to do dynamics for reaction path Hamiltonian method with as they don't have sharp crossing.
- Reducing the space of phonon modes to consider the relevant modes for dynamics.
- Diabatization of the phonon modes.
- Updating the model (in process)
- Implementing Quantum Trajectory method.

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